Jayant Chauhan

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Professional Summary

Results-driven Computer Science undergraduate specializing in Machine Learning architectures and low-level systems programming. Demonstrated impact through the recognized AI forensics platform (Top 30, India Innovates 2026) and the hackathon-winning video AI pipeline. Contributed to major open-source ecosystems (GNU Compiler Collection, GNU Octave).

Education

Netaji Subhas University of Technology (NSUT) — New Delhi, India
Bachelor of Technology in Computer Science — *Expected Graduation: 2028*

- **Relevant Coursework:** Data Structures & Algorithms, OOP (C++), Operating Systems, DBMS, Probability & Statistics, Design & Analysis of Algorithms, Computer Architecture and Organization

Work Experience

GNU Octave (Statistics Package) — Remote
Open Source Contributor (MATLAB/C++) | Statistics Package — *Oct 2025 – Jan 2026*

- **stepwisefit:** Rewrote the stepwise regression backend from scratch, implementing F-test and p-value computation for variable selection loops and Name-Value parsing via pairedArgs convention.
- **dummyvar:** Proposed and implemented one-hot encoding for numeric and categorical variables for the first time in the package; handled categories() ordering conventions, NaN-to-zero row semantics, added BISTs.

GNU Compiler Collection (GCC Rust - gccrs) — Remote
Open Source Contributor — *Jan 2026 – Mar 2026*

- **Compiler Stabilization:** Authored and merged 20+ PRs into the GNU Toolchain's Rust frontend, resolving critical Internal Compiler Errors (ICEs) in rust-attributes.cc and adding robust regression tests.
- **AST/HIR Architecture:** Developed AST and HIR reconstruction logic, creating foundational infrastructure to extend Rust pattern matching support; implemented safety checks across parse, and backend compiler passes.

Projects

Satya Setu.AI (Top 30, India Innovates 2026) | *FastAPI, XGBoost, Transformers, React*

- Designed a full-stack forensic analysis platform combining OCR (Tesseract + PyMuPDF), NER (spaCy), TF-IDF + Sentence-BERT scam classification, XGBoost-based corruption risk prediction into an evidence processing pipeline.
- Trained fraud detection model on 1.8M+ public procurement contracts (Digiwhist Romania 2023), achieving $R^2 = 0.74$ and RMSE=0.098 on a composite risk indicator target; cross-validated on international datasets for generalization.

VividReel (Hackathon Winner) | *Python, OpenCV, PySceneDetect, MoviePy, FastAPI*

- Built an end-to-end CV pipeline (PySceneDetect + OpenCV) that scores clips on focus quality, motion dynamics, and face presence, then assembles cinematic highlight reel/video via a FastAPI microservice cutting manual editing time by ~90%.
- Architected multi-stage processing with MoviePy for dynamic assembly, automated transitions, and thematic audio; deployed as a production microservice handling full event footage ingestion to final reel output.

Technical Skills

Programming Languages: Python, C++, JavaScript/TypeScript, SQL, Rust, MATLAB/Octave
ML & AI Frameworks: PyTorch, TensorFlow, Scikit-learn, Transformers, XGBoost, LangChain, Llamaindex, OpenCV
Data & Tooling: Pandas, NumPy, Jupyter, spaCy, PyScene Detect, MoviePy, Tesseract, PyMuPDF
MLOps & Engineering: FastAPI, Docker, Git/GitHub, CI/CD Pipelines, System Design, Agile/Scrum

Leadership & Volunteering

PRAYAS (NSUT) — New Delhi, India
Core Member

- Mentored underprivileged students in math and science; co-organized a seminar with Microsoft and Salesforce professionals, providing career guidance to 20+ female students.